

Vasanth Modepalli

Java Developer | 4 Years Experience
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LinkedIn | GitHub | Portfolio

PROFESSIONAL SUMMARY

Java Developer with 4 years of experience designing, building, and optimizing high-throughput backend systems using Java, Spring Boot, REST APIs, and Oracle SQL. Demonstrated expertise in backend automation, microservices, and query tuning to enhance performance. Experienced in end-to-end service ownership, dynamic integrations, and production issue resolution.

TECHNICAL SKILLS

Languages: Java 8/17, SQL, PL/SQL, Python, Kafka

Frontend: React, JavaScript (ES6+), HTML5, CSS3

Backend: Spring Boot, Spring MVC, REST APIs, Microservices, Hibernate, Spring Data JPA, FastAPI

Databases: Oracle, MySQL, PostgreSQL, Redis

Tools: Git, Postman, Maven, IntelliJ, Docker, Linux, Cron Jobs

Concepts: Backend Automation, RCA, API Integration, SQL Optimization, Data Reconciliation, Production Debugging

PROFESSIONAL EXPERIENCE

Infosys Ltd.

Jan 2024 - Present

Senior Systems Engineer — Java Full Stack Developer

- Designed and developed backend modules in Java and Spring Boot for taxpayer-facing government portal services, contributing to features used by millions of users across high-traffic filing cycles.
- Built React-based frontend forms for taxpayer data entry workflows on the income tax portal, integrating UI components with backend REST APIs for structured, validated data submissions.
- Engineered REST API enhancements and backend fixes across service and database layers — analyzed API contracts, service interactions, and DB schemas to implement correct, production-safe changes independently.
- Architected and built an automated reconciliation pipeline using Python and Oracle SQL that generates JSON payloads and triggers downstream feed processing — reducing a multi-hour manual process to under 5 minutes for 4k–5k records.
- Designed and implemented 22 dynamic email notification templates using FreeMarker-based HTML, integrating them end-to-end with backend communication services and notification dispatch logic.
- Optimized complex SQL queries and stored procedures, improving backend data retrieval performance by approximately 25% during peak filing loads — identified bottlenecks through query plan analysis and index tuning.
- Diagnosed and resolved critical production issues including feed failures, API contract mismatches, and database inconsistencies — performed structured RCA by tracing service logs, SQL execution paths, and interface payloads.
- Collaborated on backend service design decisions, reviewing data flow across microservices and ensuring consistency between API responses and downstream system expectations.

Infosys Ltd.

Nov 2022 - Jan 2024

Systems Engineer — Java Backend Developer

- Developed and maintained backend components for critical income tax portal workflows, ensuring system correctness during high-volume filing periods with concurrent user spikes.
- Designed and executed API contract validation across backend services, ensuring correct data flow and catching integration regressions before they reached production.
- Built SQL-based diagnostic utilities to trace data inconsistencies across backend services, enabling faster resolution of data integrity issues in production.
- Identified and resolved root causes of backend failures across Java services, SQL layers, and external interface payloads — reducing repeat incidents through permanent code and query-level fixes.

KEY PROJECT

Scalable URL Shortener Service

Java 17, Spring Boot, MySQL, Redis, Docker

- Designed and built a production-grade RESTful URL shortener service using Base62 encoding with collision-safe ID generation.
- Implemented a Redis caching layer with TTL-based expiry to serve high-frequency redirects with sub-millisecond latency, reducing database load significantly.
- Built an analytics module to track click counts, referrer patterns, and usage trends per shortened URL.
- Enforced input integrity using DTO-level validation and designed a centralized exception handling layer with structured error responses.
- Containerized the full application stack using Docker for environment-consistent local and deployment setup.

EDUCATION

B.Tech Mechanical Engineering

2018–2022

Sasi Institute of Technology and Engineering

CERTIFICATIONS

- Infosys Certified Spring Boot Developer
- Infosys Certified Java SE8 Developer

- Solved 150+ DSA problems on LeetCode